BHARGAV PONKIYA

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SUMMARY

Full-Stack Engineer and Tech Lead with **4+ years of experience** architecting scalable web applications, real-time voice AI systems, and cloud-native backends. Proven track record directing technical architecture from rapid proof-of-concept to production, optimizing AWS cloud infrastructure, and integrating third-party enterprise platforms. Experienced in leading international technical walkthroughs and shipping systems with **TypeScript, Node.js, NestJS, Next.js, Python, AWS**, and production LLM APIs (**OpenAI Realtime, Gemini, Claude**).

SKILLS

Languages	JavaScript, TypeScript, Python, SQL
Frontend	React.js, Next.js, Redux, Tailwind CSS, React Query
Backend	Node.js, Express.js, NestJS, GraphQL, REST APIs, tRPC, WebSockets
Databases	PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, pgvector
Cloud & DevOps	AWS (S3, EC2, CloudFront, CloudWatch, Lambda), Docker, GitHub Actions, Render, Vercel
Architecture	Microservices, Event-Driven Architecture, RabbitMQ, Nx Monorepo
AI / LLM	OpenAI Realtime API, Gemini, Claude, ElevenLabs, Deepgram, LiveKit, RAG, pgvector
Tools	Git, Postman, JMeter, Jira, WebRTC, SSE, Odoo ATS, Recruitee ATS

EXPERIENCE

GRAIXL — July 2026 – Present

Tech Lead | Remote

- Provide **technical leadership** driving end-to-end architecture, core development, and production system releases across web applications and real-time AI services.
- Architected a real-time multilingual voice AI pipeline integrating **OpenAI Realtime API, ElevenLabs, Deepgram STT/TTS**, and **LiveKit Cloud (WebRTC)**, achieving sub-second perceived latency.
- Implemented candidate video verification via **periodic random face-frame capture**; conducted conversational video AI avatar POCs with **Simli** and **ProtoFace**.
- Decoupled compute-intensive background workloads using **RabbitMQ** message queues for fully asynchronous, non-blocking processing.
- Managed AWS cloud infrastructure (**S3, CloudFront, EC2, CloudWatch**) and CI/CD automation; audited cloud assets and decommissioned idle instances to save **~$150/month**.
- Integrated **Odoo ATS** and **Recruitee ATS** via webhooks and REST APIs; conducted direct technical walkthroughs and product onboarding for international client teams.

Upsquare Technologies — Feb 2026 – Jul 2026

Software Engineer | Ahmedabad, Gujarat

SMG (Sway Me Good) – Influencer Marketing Platform | *Node.js, tRPC, GraphQL, AWS Lambda, DynamoDB, Nx Monorepo, Azure OpenAI API*

- Built *Loyalty* and *Campaign-Auth-Score* services in an **Nx monorepo (25+ microservices)**; engineered an AI authenticity scoring engine (**Azure OpenAI API**) with scheduled **AWS Lambda** jobs and **DynamoDB**.
- Designed typed API contracts using **tRPC** and **GraphQL**, optimizing data-fetching throughput across analytics dashboards.

VERSYS – Compliance & Document Management System | *Node.js, Express.js, Claude Sonnet, Puppeteer, AWS S3, React, React Query, Tailwind CSS*

- Built a centralised compliance portal and engineered an automated document generation pipeline combining **Salus API + Claude Sonnet** to render safety PDFs via **Puppeteer** to **AWS S3**; tuned **React Query** caching to eliminate redundant API requests.

EXPERIENCE (CONTINUED)

Meditab Software **Dec 2022 – Jan 2026**
Programmer Analyst | Ahmedabad, Gujarat

- Developed and maintained internal web applications (**Node.js, Express, PostgreSQL, React**) supporting QA automation workflows used daily by engineering teams.
- Integrated **Socket.io** for live script execution tracking and activity feeds, improving real-time visibility for QA teams.
- Optimised database throughput via **connection pooling** and query tuning; reduced redundant re-renders using **React.memo** and **useCallback**.
- Offloaded computational workloads using **Worker Threads**; benchmarked concurrent API performance with **JMeter** to identify and resolve throughput bottlenecks.

Codient Tech **May 2022 – Oct 2022**
Full-Stack Developer | Remote

Dausto – E-Commerce Platform | *TypeScript, Express.js, MongoDB, React + Vite (Admin), Next.js (Storefront), Formik, Zod, MSG91*

- Built core e-commerce modules across React + Vite admin panel and Next.js storefront backed by a TypeScript + Express + MongoDB API.
- Implemented **Formik + Zod** form validation and **MSG91** OTP authentication for customer onboarding and automated order notifications.

PERSONAL PROJECTS

NovaTel – AI Voice Agent with Self-Healing Prompt Pipeline | GitHub
Node.js, React, TypeScript, LiveKit, WebRTC, Deepgram STT/TTS, Gemini API, MongoDB, Socket.IO

- Built a real-time telecom billing voice agent with **Deepgram STT**, **Gemini LLM**, and **Deepgram Aura TTS**; audio transport via **LiveKit WebRTC** with **Socket.IO** control events (∼1.3s latency).
- Engineered an autonomous self-healing prompt pipeline: post-call Gemini QA rubric evaluation generates surgical patches into MongoDB, auto-loaded for subsequent calls.

Enterprise AI Agent – Cryptographic Human-in-the-Loop (HITL) Governance | GitHub
n8n, Local LLMs, HMAC Authentication, Webhooks, Docker, REST APIs

- Built a self-hosted, zero-API-cost customer support pipeline in **n8n** using **local LLMs** for high-risk Tier-2 actions (refunds, credit adjustments).
- Enforced a **cryptographic HMAC-signed HITL approval gate** preventing unauthorized financial execution or AI hallucinations.

PulseBoard – AI-Powered Developer Observability Platform | GitHub
NestJS Microservices, Next.js, MongoDB, PostgreSQL, Redis, RabbitMQ, SSE, Gemini API

- Open-source Datadog alternative correlating downtime with GitHub commits and error logs; generates automated root-cause reports via **Gemini Flash**.
- **4-service NestJS monorepo** over **RabbitMQ**; semantic log search via **MongoDB Atlas Vector Search (HNSW)**; **Redis SETNX** alert debouncing.

EDUCATION

Parul Institute of Technology 2019 – 2023
B.Tech in Computer Science and Engineering CGPA: 8.2 / 10.0

LANGUAGES

English (Fluent) | Gujarati (Native) | Hindi (Fluent)